

Geospatial proposal readiness

Not yet independently implementable

The proposal is not yet independently implementable. This loop audits all 31 functional requirements and identifies ten open categories in the data model and normative runtime behavior. Most intended outcomes are already required; their missing representation, conventions and guarantees cannot be supplied by prototype choices. Proposal readiness and requirements-build completion both remain false.

The audit and corrective experimental delivery are complete. This is not a claim that the proposal-conformance build passed. The explicitly authorized experiment executed 80 tests, 28 three-consumer jobs and 7 fresh-reader exports. It cannot close a shared semantic decision. Source `69e4cdbed03cec5a9c89b1e8938c028001e2dcf0` and every executed input are hash-bound to the receipt.

The proposal still needs a shared contract

The agreed outcomes remain useful: complete WKT CRS definitions, separate model placement, ordinary USD transforms after resolution, preserved source data and both rendering and coordinate queries. The following contracts must be completed against those outcomes rather than copied from a convenient implementation.

Gap	Shared contract to settle	Existing requirement numbers
G01	Binding carrier and contradictory legacy text	5, 6, 7, 9, 11, 19, 25
G02	Placement types, bases, defaults and frame meaning	9, 10, 11, 12, 14, 15, 20
G03	Working context and adjusted measurement placement	8, 9, 11, 16, 19
G04	Axis conventions and supported result domains	12, 13, 14, 16, 17, 18
G05	OGC-grounded WKT string normal form	1, 2, 3, 21, 27, 28, 29, 31

Existing functional intent → complete the missing model and runtime contract before dependent derivation.

The remaining contracts span data and results

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Gap	Shared contract to settle	Existing requirement numbers
G06	Measurement property association and adapters	5, 6, 7, 12, 18, 19, 21, 27, 30
G07	Complete-asset declaration and assembly obligations	7, 19, 25, 26, 27
G08	Extent/error guarantee and comparable operations	17, 18, 21, 22, 23, 24, 28, 29
G09	Structured export sampling and preservation coverage	2, 18, 19, 20, 27, 30
G10	2D/3D coordinate meaning and height-frame support	1, 4, 12, 21, 22, 30

Existing functional intent → complete the missing model and runtime contract before dependent derivation.

Why I did not stop during implementation

I substituted the frozen experimental candidate for shared-proposal authority and audited code against that candidate more thoroughly than the candidate against the actual proposal. Selecting only the requirements hid conflicting retained translate/reset/API sketches instead of reconciling the whole text.

01

I also failed to make known gaps block completion.

02

The earlier audit recorded the R24 extent/error gap, but the README opened with a complete experimental run and the slides led with a render.

03

Self-selected tests omitted bounds, relative frames and default-output behavior, and fixture-named export checks missed dropped properties.

The default runner now stops before dependent execution while semantic gaps remain. An explicit experimental invocation records `proposal_ready=false` and `requirements_build_complete=false`; there is no successful-run override that changes those conclusions.

Repairs remove shortcuts, not design decisions

A separate local proposal draft removes obsolete callable signatures, translate-based CRS placement, the helper that authored `resetXformStack` and rendering architecture prescriptions. It preserves the recorded separate-placement and post-transform direction and leaves the exact binding/field contract open. That proposal cleanup is committed locally, not pushed or agreed; its revision is recorded in the quality receipt.

Existing failure/source-preservation requirements justify rejecting a broken enclosing binding instead of treating it as absent, validating inherited coordinate roles, retaining arbitrary ordinary dataset property names/metadata/relationships, and avoiding duplicated output WKT in export metadata. The experiment also tests composed-default output precedence, affine leaf bounds including curve widths, relative model frames and standard USD instance export. Where a new experimental representation was necessary, it was specified and committed before affected code; it remains unapproved.

WKT string normalization now covers structural parentheses as well as square brackets while preserving quoted punctuation. OGC permits the former and prefers the latter; the installed projection engine rejects the parenthesis spelling directly. Parser capability therefore cannot define all OGC-valid text. The full lexical normalization profile remains G05, and lexical token identity is not semantic CRS equivalence. OGC 18-010r11, clauses 6.3.4 and 6.4.

Placed measurements expose an unresolved dependency

Absolute sample coordinates and a model's local geometry are different kinds of data. The proposal does not yet fully define how model placement and ordinary project adjustments affect arrays of absolute measurement coordinates, including the working frame and pivot when a geographic context is involved.

The city/global fixtures used unadjusted datasets. Converting those arrays demonstrates coordinate queries and preserves values/indices/times, but does not demonstrate adjusted imagery under requirements 8 and 17. I stopped that dependent derivation and recorded it in G03/G06; no centroid, hidden origin, extra flag or new placement field was introduced to rescue it.

A related distinction is CRS definition validity versus the support needed for a requested result. A valid 2D definition does not supply physical height; geographic gravity-related height cannot silently become ellipsoidal Cartesian height. The three-component candidate does not establish universal 2D/compound-height model support. G10 requires result-specific information and failure semantics, not a schema restriction copied from vector3 code.

A sampled error is not an extent guarantee

Double anchors avoid absolute-coordinate float quantization, but an affine model frame approximates nonlinear per-point conversion. Requirement 24 calls for a stated placement error over a stated extent; the proposal has not yet supplied a complete contract for establishing that guarantee, its coverage or failure/subdivision behavior.

Sampled domain half-extent	Points	Maximum affine discrepancy
±1 m	289	0.160 μm
±10 m	289	15.7 μm
±100 m	289	1.569 mm
±1,000 m	289	0.1568 m
±10,000 m	289	15.68 m
±100,000 m	289	1568 m

These are finite sampled discrepancies, not a bound on every point or surface in the extent. The ±100 km sample is about 1.57 km from the per-point result.

The added affine leaf-bounds controls use stock UsdGeom local extent, including curve widths, transformed by the same derived frame.

One authored dataset, three runtime results

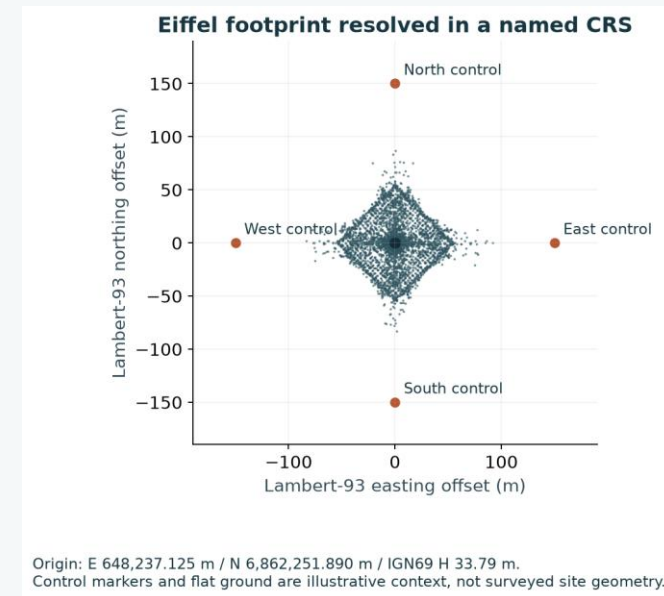
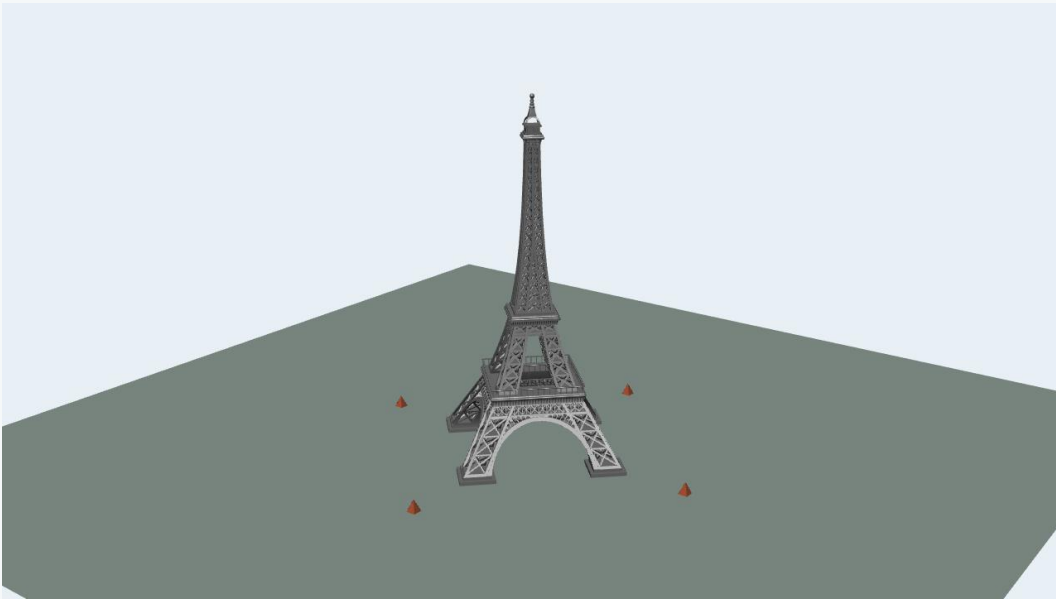
The corrective experiment compares headless OpenUSD queries, an independent native Hydra resolver and an independent live OV resolver on the same authored data. It records 760,792 positional samples across 28 jobs, 51 affine leaf-bound comparisons and 76 relative-frame comparisons. Maximum native discrepancy is 0.000104 mm and maximum OV discrepancy is 0.000000 mm.

Same first Colorado breakline vertex	Easting (m)	Northing (m)	Height (m)
Headless USD	490154.882832	4416473.740389	1655.208918
Native Hydra resolver	490154.882832	4416473.740389	1655.208918
Live OV resolver	490154.882832	4416473.740389	1655.208918

All three consumers use PROJ, and the supplied partner CSVs also use PROJ. This is independent authored-scene/runtime agreement, not independent geodetic-engine accuracy. Actual final Hydra matrices, instance transforms and extents are read back; live OV checks ingested geometry, widths, runtime matrices and coordinate results. No OV visual-render parity is claimed.

Eiffel: see the model and its coordinate context

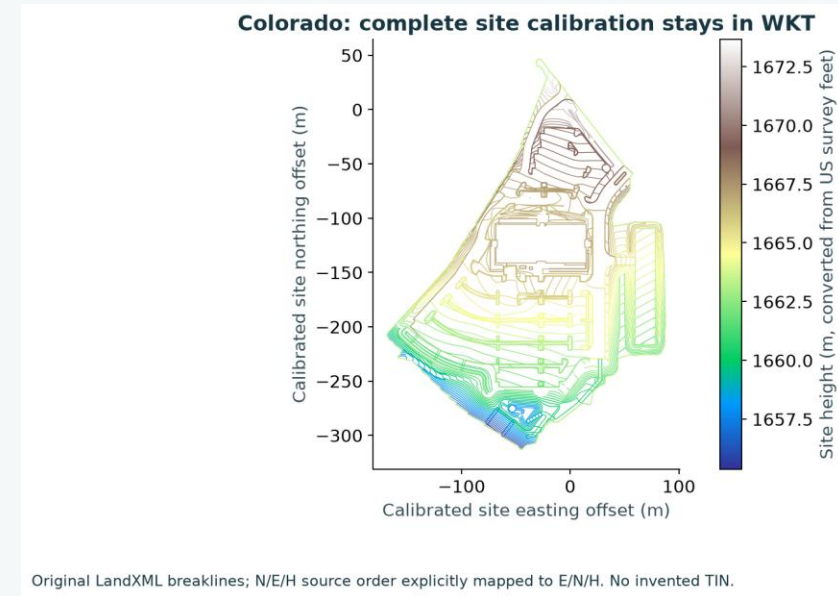
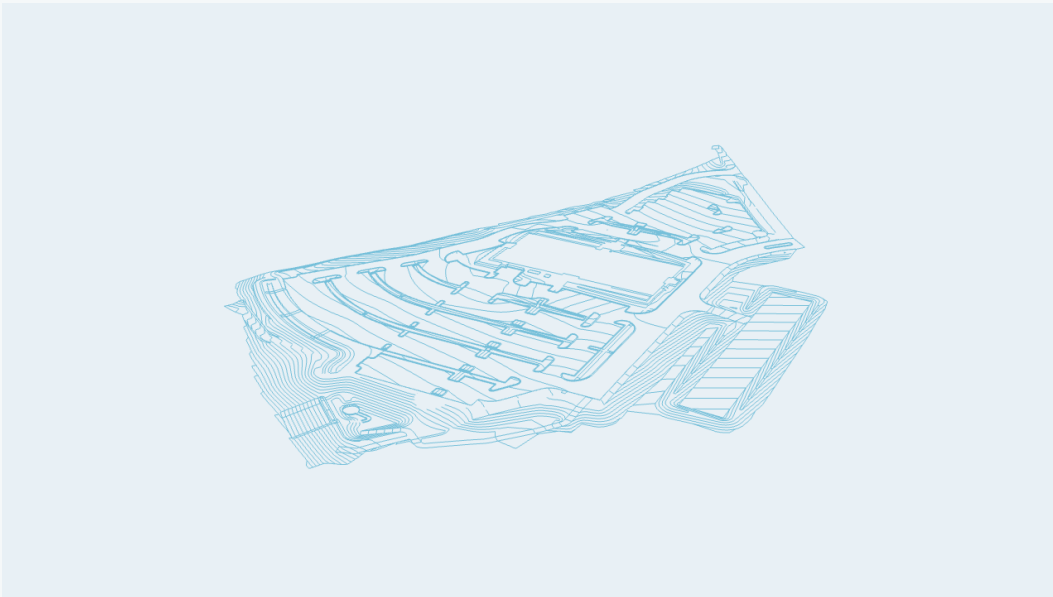
The Eiffel model is placed at Lambert-93 E 648,237.125 m, N 6,862,251.890 m and IGN69 height 33.79 m, with a 45° CRS orientation. The same authored model resolves into Lambert-93, two UTM zones and ECEF.



The first image is an actual native Hydra/Storm draw from the early candidate scene-index override, before instance propagation and flattening. The second is a headless query plot in named coordinate axes. Ground and cardinal controls are illustrative, not a surveyed Paris dataset. Both images are needed: appearance alone is insufficient evidence of correct placement.

Complete CRS definitions support calibrated sites

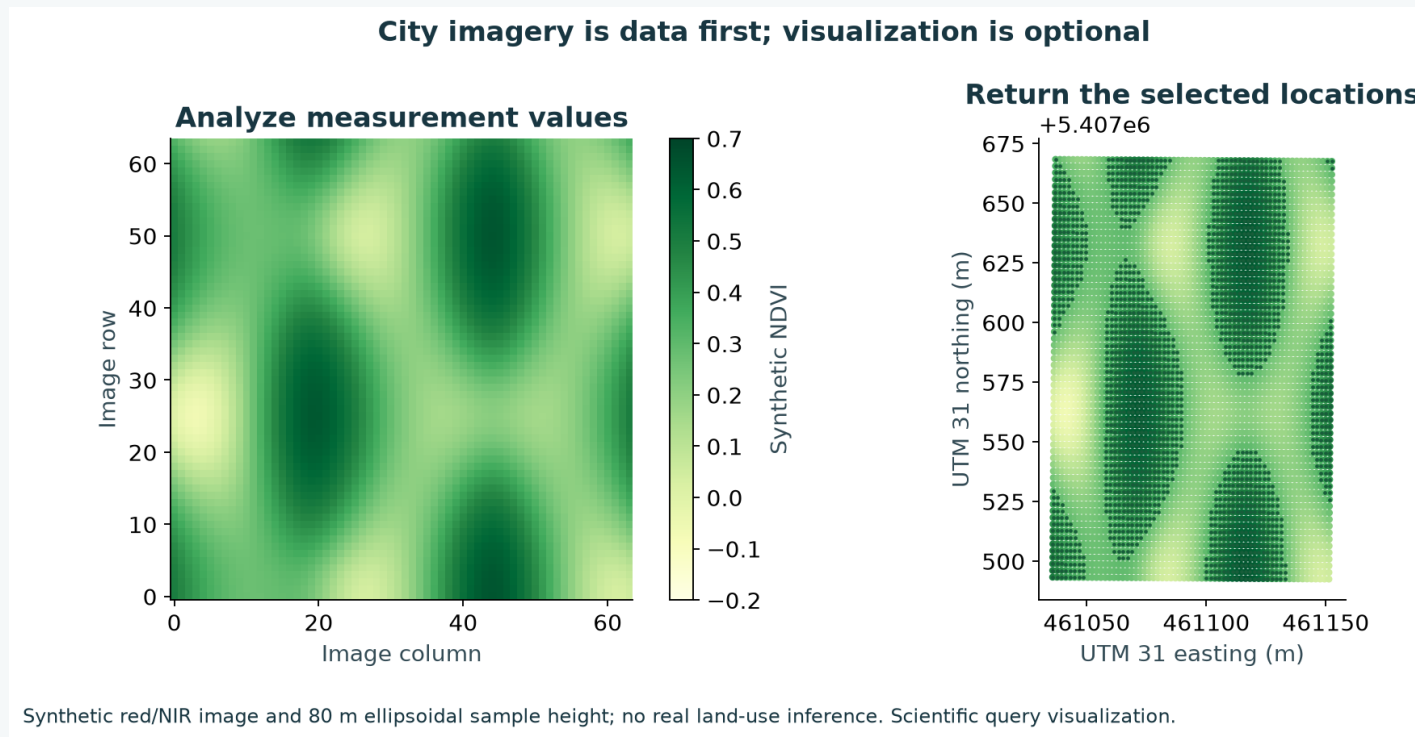
The partner data exercise geographic, projected, compound height and derived site-calibration definitions. Colorado keeps US survey feet and its full horizontal/vertical site calibration in WKT; France keeps Lambert-93/CC49 and IGN69 height meaning.



The LandXML contains 14,359 vertices in 710 coordinate parts. It is displayed as original breaklines, without inventing a terrain TIN. Source N/E/H order is explicitly mapped to E/N/H, with metre local geometry and a double coordinate anchor. Five Colorado and 59 France controls are checked in their supplied reference systems.

City queries produce reusable analytic data

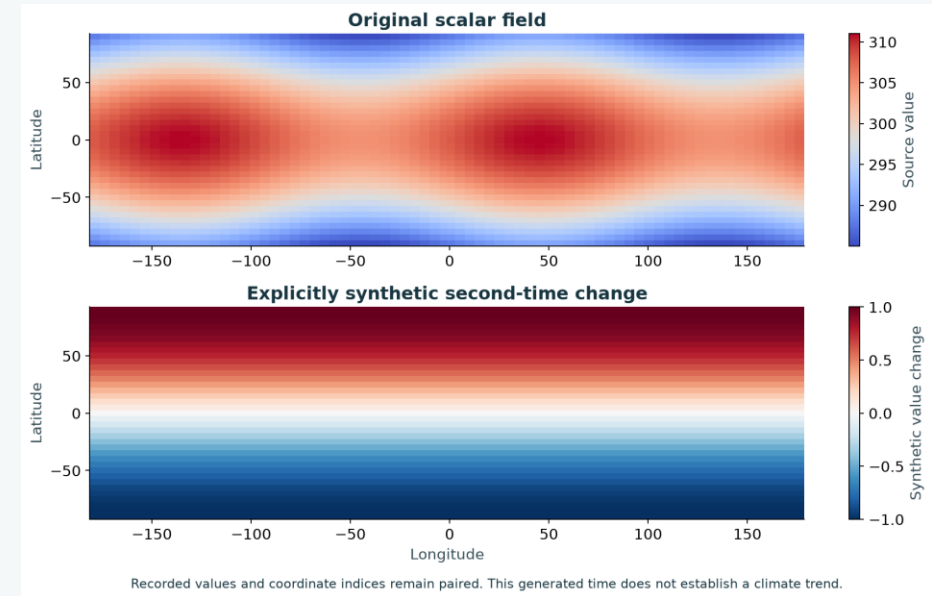
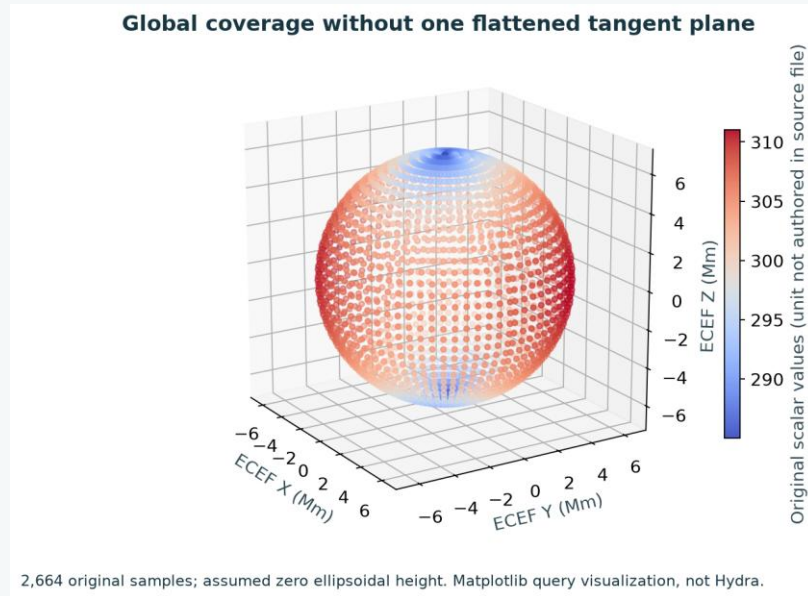
A 64×64 synthetic red/NIR image yields 1,850 samples above NDVI 0.3. The result returns locations in the requested CRS while preserving source indices, values and observation times.



The heatmap is a scientific headless-query visualization. Its bands, observation time and 80 m ellipsoidal sample height are explicit synthetic inputs; it makes no real land-use claim.

Global data retains coordinates, values and time

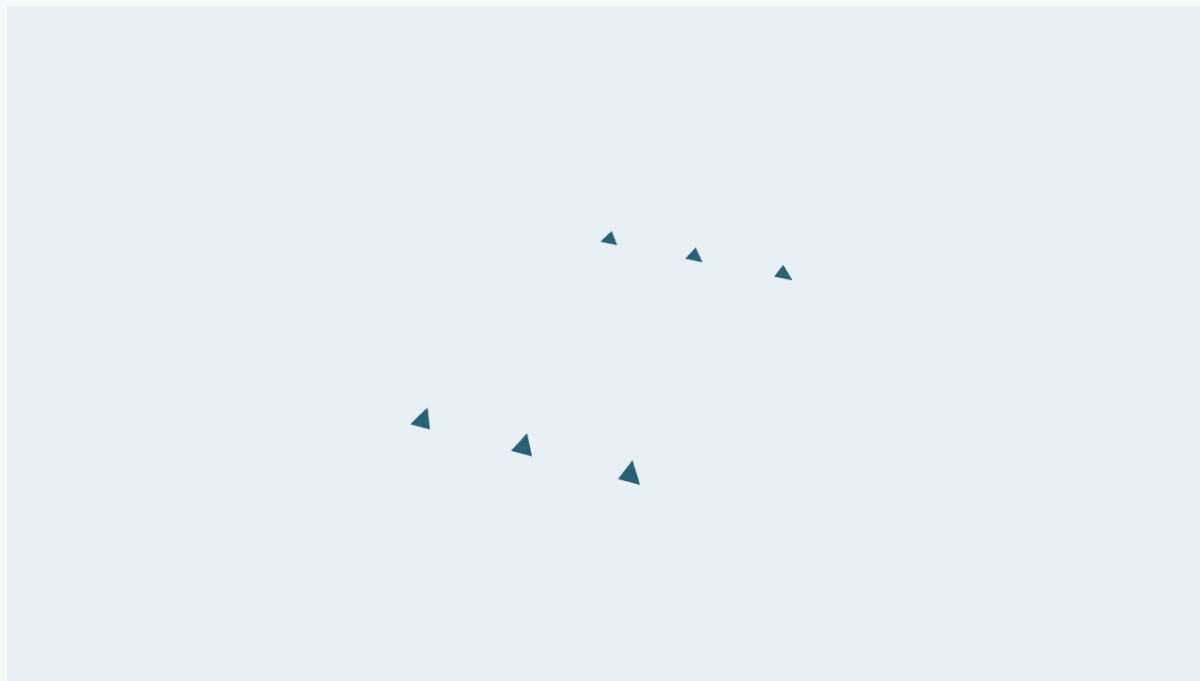
The original 37×72 scalar grid resolves all 2,664 locations globally in ECEF and geographic coordinates. Values remain associated with their coordinates and grid indices across two recorded times.



The initial scalar field comes from the supplied NetCDF; its units and heights are not authored in that file. Positions use explicitly assumed zero ellipsoidal height. The second time is an explicitly synthetic perturbation, not an observed trend or forecast. The analytic product preserves both value arrays with the same coordinate indices. These are scientific query plots, not Hydra renders and not one global tangent plane.

Real consumer edits and fresh-reader exports

Three native USD instances and three point instances share ordinary geometry while resolving distinct placements. References, stronger opinions, variants, source interpolation, stage units/up-axis and explicit/default outputs have distinguishing controls. Native source notices and live OV source/request edits invalidate results; broken definitions remove stale successful results and recovery is exercised.



7 derived assets are reopened and resolved at explicit times 0, 5 and 10.

This is an experimental export representation, not an approved standard encoding.

Proposal closure governs the next derivation

The next shared revision must reconcile the binding and placement model, define adjusted measurement/frame semantics and result support, prescribe WKT string normalization, and complete dependency, export and extent/operation guarantees. Every repair must cite existing requirements and USD/OGC semantics before affected implementation is rederived. Candidate choices and test success cannot settle those decisions.

Coordinate epochs remain deferred in representation, interpretation and computation. CRS frame epochs remain in WKT, and supported non-epoch datum/height operations use engine-managed resources. Large city/global coordinate datasets remain useful initial scope; the construction examples do not impose a site-sized domain. No deferred capability is made impossible by a hidden surrogate field.

The full requirement/gap map and why-stop investigation are in `QUALITY_REVIEW.md`. The run delivers auditable evidence and current collateral while honestly failing the proposal-readiness gate.